Understanding Imputation of Missing Values

by Shahriar Analytical Academy

What is Imputation of Missing Values?

Definition

Imputation is the process of filling in missing data points with estimated values instead of deleting them. It allows us to retain as much information as possible in the dataset while reducing bias and maintaining data integrity.

Purpose of Imputation

- To preserve sample size
- To avoid biased results due to non-random missingness
- To allow statistical and machine learning models to work properly

Real-Life Analogy: Imagine a survey where some participants forgot to write their age. If you remove those surveys, you lose valuable feedback. But if you intelligently estimate their age based on similar participants, you keep the insights intact — that's imputation.

Why Imputation Is Better Than Deletion

1 **Deletion Removes Data Permanently**

Losing data means losing statistical power.

Can introduce bias if data isn't missing completely at random (MCAR).

2 **Imputation Preserves Patterns**

It helps maintain the structure and distribution of the dataset.

Advanced imputation uses relationships between variables.

3 **Real-World Data Is Rarely Perfect**

In business, healthcare, or social research, deleting rows with missing values could mean throwing away crucial insights.

Types of Imputation Techniques

Simple Imputation

1

- Mean Imputation
- Median Imputation
- Mode Imputation

Custom Domain-Based Imputation

2

Based on domain logic (e.g., if gender is male, estimate height differently)

Simple Imputation Methods

1

Mean Imputation (for numeric data)

Replace missing values with the column's mean. Best when data is normally distributed.

2

Median Imputation (for numeric data)

Use the column's median instead. Best when data is skewed or has outliers.

3

Mode Imputation (for categorical data)

Use the most frequent category. Good for nominal variables like gender, region, etc.

Key Learning Points

1

Retain Valuable Data

Imputation helps you retain valuable data instead of discarding it.

2

Choose Method Wisely

Not all missing values should be handled the same way. Choose a method based on data type and structure.

3

Consider Technique Complexity

Simple imputation is fast but limited. Advanced techniques use inter-variable relationships.

4

Analyze Missingness Patterns

Always analyze the pattern of missingness before deciding on an imputation strategy.

5

Improve Model Robustness

Proper imputation leads to more robust models and insights.

Thank You!

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